Design Milestones

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IDE 402.A

Concept Development

Initial Concept Exploration

Precision
Semi-automatic
Ritual

Prompt
A home barista machine — semi-automatic with a built-in grinder, steamer, and espresso extraction. It can be used in offices and homes. The machine should not limit users, they still control grind size and temperature.

Design Decision
At this stage, the concept focused on integrating the entire espresso preparation workflow into one machine.
The goal was to reduce the need for multiple devices while maintaining the authentic steps of espresso preparation.

What I Transformed
The integrated espresso station concept was too complex and multifunctional.
Through later iterations, the project shifted toward specializing in one stage of the process: milk preparation, which led to the final concept of a dedicated milk texturing device.

AI Role
Clarifier + Expander
AI helped structure the early concept by defining the user profile and use context.

Alternative Concepts

01 Adaptive Milk Heater / Steamer

A dedicated milk heating & steaming device that adapts to different milk types and drink intentions, supporting everyday beverage rituals through clear and sensory interactions.

Isolates milk preparation as a core ritual, offering precise results while allowing users to complete their beverages independently.

02 Integrated Milk & Infusion Maker

An all-in-one hot beverage system that prepares milk and dry ingredients together, translating diverse drink traditions into a simple, guided, and inclusive preparation experience.

Prepares both milk and ingredients together, translating it into a single process.

- *increased technical complexity and reduced user involvement*

- Froth / steam milk with precision
- Consistent foam from various milk types
- Temperature and foam level control

03 Guided Beverage Making Station

A guided countertop platform that structures hot beverage preparation as a legible ritual through stages, pacing, and sensory feedback rather than fixed automation.

Structures the entire preparation as a shared ritual, using guidance and pacing to support understanding and awareness rather than automated execution.

My Design Decision

Focus the product on milk preparation only, leaving beverage preparation to the user.

Accepted

The framing of one device that does one thing well.

Rejected

Fully automated drink preparation.

Why This Direction?

This direction was selected because milk preparation is a frequently inconsistent part of hot beverage making that is currently underserved by existing machines. By focusing on milk as a primary system, the design can support multiple beverages, users and contexts while directly addressing quality and usability without competing with traditional coffee machine typologies.

Impactful Prompts

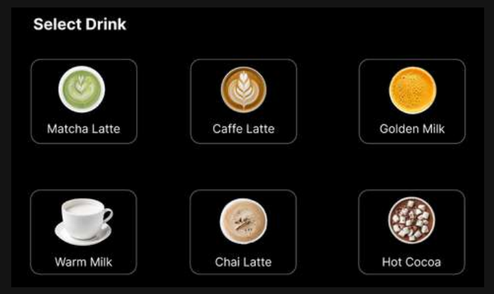
A countertop device dedicated to heating and texturing milk correctly for different beverages, adapting its behavior based on milk type and intended drink, while keeping the experience simple and calm. Rather than automating the entire drink, the machine focuses on doing one thing exceptionally well: preparing milk with quality and consistency.

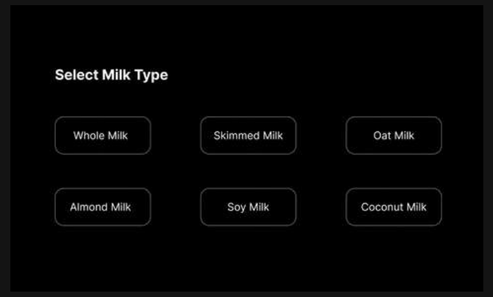
Create an image of a milk centered countertop device that prepares milk correctly for different hot beverages and milk types, shaping heating and steaming into a clear, adaptive ritual rather than an automated task.
What the device does: Heats and steams milk with precision
Adapts behavior based on milk type (dairy, oat, soy, almond...)
Preparation intent (amount of foam)

Ideation & Prototyping

Safety states
Process
Interaction clarity

Simple 4 step digital interface prototype





Application

My Reflection

This helped shift my thinking from object design to system behavior.

My Design Decision

Design the interface around a few meaningful inputs rather than complex settings.

AI Role

Re-framer

Accepted

Structured system thinking.

Rejected

Too many user controls & personal settings.

Mechanical Concept

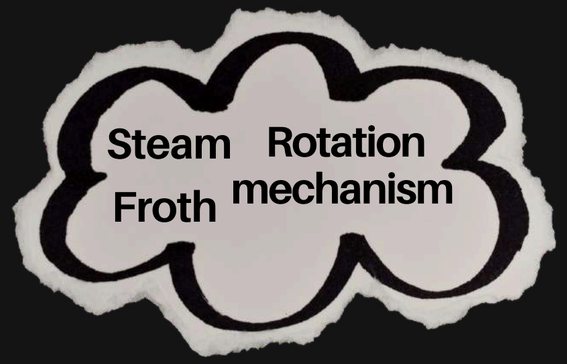
My Design Decision

Use two separate functions:

Steam → heating

Froth → aeration

Both rotate from a circular mount above the pitcher.



AI Role

Expander

Accepted

Dual-wand mechanism..

Rejected

Hidden internal frothing systems

Selected Prompt

"Create a render showing the milk frothing mechanism."

Process Execution

Steam wand injects steam for heating

Frothing spindle spins to incorporate air and texture the milk

If frothing is required

The plate rotates to position the frothing spindle above the pitcher.

If steaming is required

The plate rotates to align the steam wand.

Form & Product Visualization

Selected Prompt
"Create a realistic render of a countertop kitchen appliance that steams & froths milk based on the drink selected by the user. the milk jug is stainless steel and the machine is shiny white/creme with steel accents."

AI Role
Critique + Expander

AI generated visual concepts that helped explore:
Wand positioning
Pitcher integration
Material combinations

My Design Decision

Final design features:

- A clear milk pitcher integrated with the machine
- A visible water reservoir
- Two rotating wands
- Minimal digital interface

Accepted

Minimal design with simple interface.

Rejected

Overly futuristic, large or complex designs.

My Reflection

AI accelerated visual exploration but required continuous correction and direction.

Thank You!